SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 50%

QUESTIONS: 5 Total Marks:50 Annexure A

Experiments

Criteria	Understanding of Concepts ***(2.5)***	Experimental Design ***(3)***	Use of Tools***(2)***	Analysis of Results***(1.5)***	Documentation ***(1)***	***Total(10)***
Weightage	*25%*	30%	20%	15%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I A. LOKESH KUMAR(192524157) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Annexure C

Experiments:

Children of three ages are asked to indicate their preference for three photographs of adults. Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study? (10 Marks)

Photograph:

Age of child A B C

5-6 years: 18 22 20

7-8 years: 2 28 40

9-10 years: 20 10 40

Use cov() to calculate the sample covariance between B and C.

Use another call to cov() to calculate the sample covariance matrix for the preferences.

Use cor() to calculate the sample correlation between B and C.

Use another call to cor() to calculate the sample correlation matrix for the preferences.

2. Assume the Tennis coach wants to determine if any of his team players are scoring outliers. To visualize the distribution of points scored by his players, then how can he decide to develop the box plot? (10 Marks)

Player ID Player Name Points Scored

P1 Player A 12

P2 Player B 15

P3 Player C 14

P4 Player D 10

P5 Player E 18

P6 Player F 20

P7 Player G 22

P8 Player H 13

P9 Player I 11

P10 Player J 35

P11 Player K 16

P12 Player L 17

Player ID Player Name Points Scored

P13 Player M 19

P14 Player N 21

P15 Player O 23

3. A bank wants to decide whether to approve a loan based on customer attributes like:

- Age
- Income
- Employment Status
- Credit Score

Using historical data, build a **Decision Tree classifier** to predict **Loan Approval (Yes/No)**. (10 Marks)

Consider the Data Set
young,high,employed,good,yes
young,high,self-employed,average,yes
middle,high,employed,good,yes
old,medium,employed,good,yes
old,low,unemployed,poor,no
old,low,self-employed,average,no
middle,low,unemployed,poor,no
young,medium,employed,average,yes
young,low,unemployed,poor,no
old,medium,self-employed,good,yes
middle,medium,employed,average,yes
middle,high,self-employed,good,yes
old,medium,unemployed,poor,no

young,medium,self-employed,average,yes
middle,low,employed,poor,no

4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: (5 Marks)
Class A: 76, 35, 47, 64, 95, 66, 89, 36, 8476,35,47,64,95,66,89,36,84
Class B: 51, 56, 84, 60, 59, 70, 63, 66, 5051,56,84,60,59,70,63,66,50
(i) Find which class had scored higher mean, median and range.
(ii) Plot above in boxplot and give the inferences

5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results: (10 Marks)

age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
age	52	54	54	56	57	58	58	60	61
%fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

(a) Calculate the mean, median, and standard deviation of *age* and
%fat. (b) Draw the boxplots for *age* and *%fat*.
(c) Draw a *scatter plot* and a *q-q plot* based on these two variables.

Experiment 1:

```
> # Create data
> data <- data.frame(
+   A = c(18, 2, 20),
+   B = c(22, 28, 10),
+   C = c(20, 40, 40)
+ )
>
> # Display data
> print(data)
   A  B  C
1 18 22 20
2  2 28 40
3 20 10 40
>
> # Chi-square test for age and photograph preference
> table <- matrix(c(18,22,20,
+                   2,28,40,
+                   20,10,40),
+                 nrow = 3, byrow = TRUE)
>
> chisq.test(table)

        Pearson's Chi-squared test

data:  table
X-squared = 29.603, df = 4, p-value = 5.895e-06

>
> # Covariance between B and C
> cov(data$B, data$C)
[1] -20
>
> # Covariance matrix
> cov(data)
          A   B         C
A  97.33333 -74 -46.66667
B -74.00000  84 -20.00000
C -46.66667 -20 133.33333
>
> # Correlation between B and C
> cor(data$B, data$C)
[1] -0.1889822
>
> # Correlation matrix
> cor(data)
           A          B          C
A  1.0000000 -0.8183918 -0.4096440
B -0.8183918  1.0000000 -0.1889822
C -0.4096440 -0.1889822  1.0000000
> |
```

Experiment 2 :

```
> # Correlation matrix
> cor(data)
           A          B          C
A  1.0000000 -0.8183918 -0.4096440
B -0.8183918  1.0000000 -0.1889822
C -0.4096440 -0.1889822  1.0000000
> # Points scored
> points <- c(12,15,14,10,18,20,22,13,11,35,16,17,19,21,23)
>
> # Summary
> summary(points)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  10.00   13.50   17.00   17.73   20.50   35.00
>
> # Boxplot
> boxplot(points,
+         main = "Boxplot of Tennis Players' Points",
+         ylab = "Points Scored")
>
> # Find outliers
> boxplot.stats(points)$out
[1] 35
>
> |
```

Experiment 3:

```
n= 15

node), split, n, loss, yval, (yprob)
      * denotes terminal node

1) root 15 6 yes (0.4000000 0.6000000) *
>
> # Plot decision tree
> rpart.plot(model,
+            main = "Decision Tree for Loan Approval")
Error in rpart.plot(model, main = "Decision Tree for Loan Approval") :
  could not find function "rpart.plot"
>
> # Predictions
> predict(model, type = "class")
  1   2   3   4   5   6   7   8   9  10  11  12  13  14  15
yes yes yes yes yes yes yes yes yes yes yes yes yes yes yes
Levels: no yes
> |
```

```
# Install package if required
# install.packages("rpart")
# install.packages("rpart.plot")

library(rpart)
library(rpart.plot)
rror in library(rpart.plot) : there is no package called 'rpart.plot'

# Create dataset
data <- data.frame(
  Age = c("young","young","middle","old","old",
          "old","middle","young","young","old",
          "middle","middle","old","young","middle"),

  Income = c("high","high","high","medium","low",
             "low","low","medium","low","medium",
             "medium","high","medium","medium","low"),

  Employment = c("employed","self-employed","employed","employed",
                 "unemployed","self-employed","unemployed",
                 "employed","unemployed","self-employed",
                 "employed","self-employed","unemployed",
                 "self-employed","employed"),

  Credit = c("good","average","good","good","poor",
             "average","poor","average","poor","good",
             "average","good","poor","average","poor"),

  Loan = c("yes","yes","yes","yes","no",
           "no","no","yes","no","yes",
           "yes","yes","no","yes","no")
)

# Convert to factors
data[] <- lapply(data, factor)

# Build decision tree
model <- rpart(
  Loan ~ Age + Income + Employment + Credit,
  data = data,
  method = "class"
)

# Display model
print(model)
= 15
```

Experiment 4 :

```
> # Class A
> A <- c(76,35,47,64,95,66,89,36,84,
+        76,35,47,64,95,66,89,36,84)
>
> # Class B
> B <- c(51,56,84,60,59,70,63,66,50,
+        51,56,84,60,59,70,63,66,50)
>
> # Mean
> mean(A)
[1] 65.77778
> mean(B)
[1] 62.11111
>
> # Median
> median(A)
[1] 66
> median(B)
[1] 60
>
> # Range
> range(A)
[1] 35 95
> range(B)
[1] 50 84
>
> # Difference between maximum and minimum
> diff(range(A))
[1] 60
> diff(range(B))
[1] 34
>
> # Boxplot
> boxplot(A, B,
+         names = c("Class A", "Class B"),
+         main = "Comparison of Class A and Class B",
+         ylab = "Exam Marks")
>
> |
```

Experiment 5

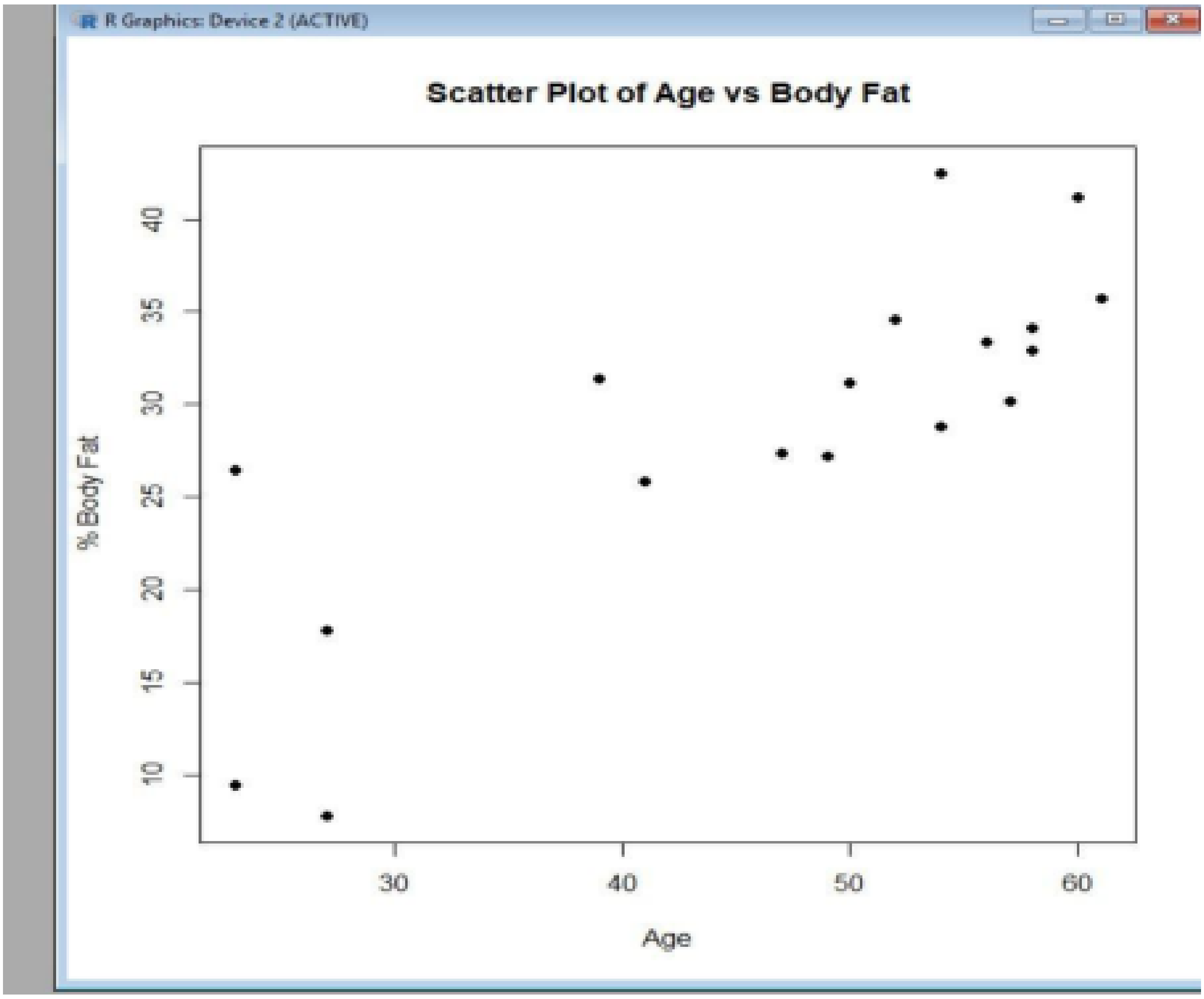
a.

```
>
> # Age data
> age <- c(23,23,27,27,39,41,47,49,50,
+          52,54,54,56,57,58,58,60,61)
>
> # Body fat data
> fat <- c(9.5,26.5,7.8,17.8,31.4,25.9,27.4,27.2,31.2,
+          34.6,42.5,28.8,33.4,30.2,34.1,32.9,41.2,35.7)
>
> # Mean
> mean(age)
[1] 46.44444
> mean(fat)
[1] 28.78333
>
> # Median
> median(age)
[1] 51
> median(fat)
[1] 30.7
>
> # Standard deviation
> sd(age)
[1] 13.21862
> sd(fat)
[1] 9.254395
>
```

b.

c.

```
plot(age, fat,
     main = "Scatter Plot of Age vs Body Fat",
     xlab = "Age",
     ylab = "% Body Fat",
     pch = 19)
```


R Graphics: Device 2 (ACTIVE)
Scatter Plot of Age vs Body Fat
% Body Fat
10
15
20
25
30
35
40
30
40
50
60
Age